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Thermo-chemical disposal of plastic waste from end-of-life vehicles (ELVs) using CO_2

Jung-Hun Kim [a,2], Sungyup Jung [b,2], Taewoo Lee [a], Yiu Fai Tsang [c,1], Eilhann E. Kwon [a,1,*]

[a] Department of Earth Resources & Environmental Engineering, Hanyang University, Seoul, 04763, Republic of Korea
[b] Department of Environmental Engineering, Kyungpook National University, Daegu, 41566, Republic of Korea
[c] Department of Science and Environmental Studies and State Key Laboratory in Marine Pollution, The Education University of Hong Kong, Tai Po, New Territories, 999077, China

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ABSTRACT

The source reduction of plastic waste could be an effective means to attenuate hazardous environmental problems triggered by microplastics. Energy recovery from plastic waste through thermochemical processes is a desirable valorization route. To realize the grand challenges, plastic waste derived from end-of-life vehicles (ELVs) was pyrolyzed. To propose a greener feature, CO_2 was introduced as a mediator to maximize carbon allocation to the gaseous pyrogenic product (syngas) by CO_2 reduction to CO and concurrent oxidation of volatile matter (VM) that was evolved from the thermolysis of plastic waste. As such, fundamental and systematic works were conducted to delineate the CO_2 effects on conversion of VMs. This study experimentally proved that CO_2 promotes thermal cracking in line with C–C bond scissions. However, the reaction rate for the conversion of CO_2 and VM into CO via homogeneous reaction was not fast. Therefore, a Ni-based catalyst was employed to accelerate the reaction rate. However, there was coke deposition on the catalyst surface. To prevent coke formation, we chose a method to enhance CO_2 reduction to CO and the oxidation of VM. Thus, three bimetallic catalysts were used for catalytic pyrolysis. Among the three bimetallic catalysts, $Rh_{0.1}Ni_1/SiO_2$ was the most effective.

1. Introduction

Plastics have been substituted for wood, metal, and ceramics owing to their versatile physicochemical features (they are lightweight, durable, and non-corrosive) [1]. The mass production of plastics, in line with their low production costs, has catalyzed their commercial/practical use in diverse applications [2,3]. Therefore, the world plastic generation in 2019 was 460 million tons [4]. Owing to the short life cycle of plastics, their use has led to enormous waste generation [2,5]. The total amount of plastic waste production was more than doubled over the last two decades [4].

Plastic waste landfilling has long been performed as a conventional waste treatment process [6]. A lot of plastic waste (60 wt%) has been landfilled without an appropriate treatment [4]. Indeed, plastic waste landfilling cannot be a sustainable disposal platform because of the negligible biodegradability of synthetic plastic materials [7]. For example, plastics can persist for several centuries in ecosystems [8]. The recalcitrant presence of plastics in landfill sites could be a specific source to release micro-sized plastics (MPs) into the ecosystem [9]. MPs are effective vectors that disturb the fate of (in)organic waste materials owing to their hydrophobicity [10]. In addition, the invasion of MPs to the living organisms poses the hazardous risk of accumulating MPs in complex food webs, resulting in adverse effects on human health [11]. Specific additives in plastics can also be released via natural or complex degradation or weathering steps [12,13]. Nonetheless, the low technical readiness of analytical techniques for quantifying the mass of MPs makes it difficult to establish a disposal platform for MPs [14]. The removal efficiency of MPs cannot be determined without quantifying their absolute masses in all environmental media [15]. Thus, the source reduction of plastic waste could be the only strategic/precautionary measure to attenuate the unwanted discharge of MPs into our ecosystem.

* Corresponding author.
E-mail address: ek2148@hanyang.ac.kr (E.E. Kwon).
[1] The two authors were equally contributed to this work as corresponding authors.
[2] These authors contributed equally to this work as co-first authors.

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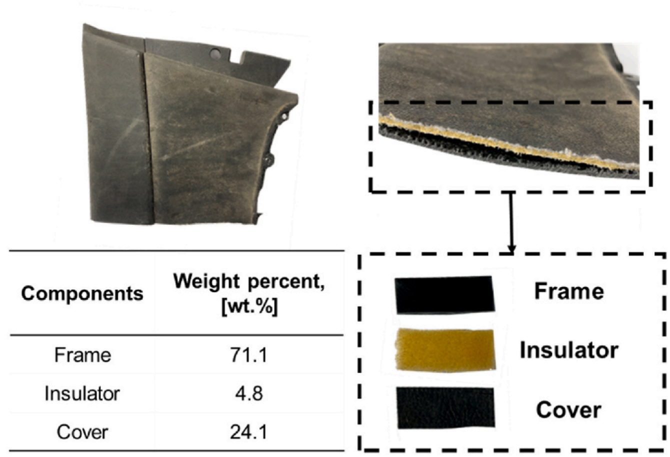


Components	Weight percent, [wt.%]
Frame	71.1
Insulator	4.8
Cover	24.1

Fig. 1. Three components in the waste door panel (WDP) from end-of-life vehicles (ELVs).

The incineration of plastic waste could be a viable technical choice, resulting in complete thermal degradation, concurrently generating heat energy [16]. The heating value of plastics is similar to that of diesel and gasoline fuels [17]. Nevertheless, plastics cannot be completely incinerated because complete oxidation of long chain hydrocarbons is technically challenging in controlling the mixing ratio of air to plastics [18]. This suggests that the fate of combustion byproducts [CO, unburned hydrocarbons (UHCs)], polycyclic aromatic hydrocarbons (PAHs), and hazardous waste components such as dioxins cannot be properly controlled [19,20]. To increase the combustion efficiency, it is of great importance that solid-phase plastic wastes are transformed into liquid/gaseous hydrocarbons (HCs) [21]. Indeed, the combustion of liquid/gaseous HCs offers a favorable chance to meet air quality standards owing to the easy/effective control of the mixing ratio of air to fuel [22]. Thus, the pyrolysis of plastic waste could be a viable technical option because solid-phase plastic waste turns into liquid/gaseous HCs [23]. In addition, char formation from the pyrolysis of broadly used thermoplastics [polyethylene (PE), polypropylene (PP), and polystyrene (PS)] is negligible [24].

Controlling the carbon chain length of liquid HCs (oil) derived from thermal degradation of plastic is a significant challenge [25]. Additional unit operations (such as separation/purification) are also required to produce HCs with the desired carbon length ($\leq C_{25}$) [26]. These processes have been perceived as critical constraints that reduce economic viability [27]. Thus, the direct transformation of solid-phase plastic waste into syngas could offer a strategic principle for environmentally sound disposal and valorization. Syngas can be efficiently combusted at a controllable equivalent ratio [28]. Additionally, the high reactivity of syngas enables its use as a raw material in the synthesis of value-added chemicals [29]. Based on this rationale, the ultimate goal of this study was to directly convert plastic waste into syngas with suppression of catalyst deactivation. For a realistic study, a waste vehicle door panel (WDP) [collected from end-of-life vehicles (ELVs)] was selected. Except for tires, PP accounted for the largest mass portion (32 wt%) in plastics used in a vehicle [30]. To impart sustainable and resilient features to the direct conversion (pyrolysis) of WDP into syngas, this work introduced CO_2 as a promoter for syngas generation. In our previous studies, CO_2 enhanced syngas formation [31,32]. Specifically, the catalytic pyrolysis (CatPyro) in addition of Ni/SiO_2 catalyst and CO_2 gas highly contributed to rapid transformation of PP into syngas [33]. However, catalyst deactivation by coke deposition on Ni/SiO_2 has not yet been resolved in a fully transparent manner. To suppress coke deposition and improve syngas production performance, Ni-based bimetallic catalysts were tested in this study. Three noble metals (Pt, Pd, and Rh) were used to synthesize the bimetallic catalysts. These noble metals were selected to prepare bimetallic catalysts, because they were known to suppress carbon decomposition and produce carbon monoxide during dry reforming of hydrocarbons in the presence of CO_2 [34]. To the best of our knowledge, the use of bimetallic catalysts for the direct conversion of WDP and CO_2 into syngas has not yet been reported. Thus, the performance of WDP on CatPyro was investigated over Ni catalyst and different bimetallic catalysts in a CO_2 environment.

2. Materials and methods

2.1. Samples and chemicals

WDP was provided by a local automobile shredding facility in Seoul, South Korea. The dust in the WDP was removed by blowing air. The three components (frame, cover, and insulator) of WDP are delineated in Fig. 1. The masses of the three components are shown in Fig. 1. The three components of WDP were dried at 80 °C for 1 d. The three dried components of the WDP were shredded and homogenized into millimeter-sized fragments. PP (Prod. #428116), silica gel (Prod. #60741), palladium nitrate dihydrate (Prod. #76070), rhodium nitrate hydrate (Prod. #83750), tetraammineplatinum nitrate (Prod. #482293), nickel nitrate hexahydrate (Prod. #203874), methylenediphenyl diisocyanate (Prod. # 256439), poly(ethylene glycol) (Prod. #8.07483), poly(vinyl chloide) (Prod. #389293), nitric acid (Prod. #438073), acetone (Prod. #34850), and dichloromethane (Prod. #34856) were provided from Sigma-Aldrich (USA). N_2, CO_2, and air were obtained from Green Gas (South Korea). N_2 balanced standard gas mixture (20 vol% H_2, 20 vol% CH_4, 20 vol% CO, and 20 vol% CO_2) was custom-made and supplied by RIGAS (South Korea).

2.2. Sample characterization

Fourier transform infrared (FT-IR) spectrometry (Agilent Cary 630, USA) was used to seek the exact types of plastics (the three components). Solvent extraction of the three components was performed to obtain specific information on additives such as phthalates. Each sample (50 mg) was mixed with 10 mL of acetone for 1 d at 20 °C. The chemical species extracted from the three components were determined using a gas chromatograph connected to a mass spectrometer (GC/MS: Agilent 5977 B, USA). Ultimate analysis was performed to determine the organic content in the WDP (Elementar Analysensysteme GmbH, Germany). The elemental compositions of the frame, insulator, cover, and WDP are listed in Table S1. The inorganic (metal) content in the WDP was determined using spectrophotometry with an inductively coupled plasma optical emission spectrometer (ICP-OES; Agilent 5800, USA). A microwave digester was used to pretreat the WDP. The WDP was mixed with nitric acid (20:1 by w/v%), and the mixture was treated using a microwave digester at 1000 W (180 °C for 20 min). Thermo-gravimetric analysis (TGA: Netzsch STA4495F5 Jupiter, Germany) was performed to monitor the thermolytic patterns of WDP under the different thermolytic temperatures (from 35 °C to 900 °C at a heating rate of 10 K min^{-1}) and different atmospheric conditions. A flow rate of atmospheric gases was 100 mL min^{-1}.

2.3. Catalyst synthesis

A monometallic Ni catalyst was dispersed on an SiO_2 support (Ni/SiO_2) using an impregnation method mentioned elsewhere [35]. A Ni precursor in DI water was prepared and added to the SiO_2 support in drops. The Ni precursor loaded onto the SiO_2 was dried at 105 °C for 12 h. The dried sample was then placed in a furnace for calcination at 600 °C for 4 h in air. The calcination temperature increased by 600 °C with a heating rate of 2 K min^{-1}. After calcination, the temperature of the prepared catalyst was reduced to 600 °C for 4 h with 20 vol% H_2. The calcination and reduction temperature programs were identical. To synthesize the bimetallic Ni catalysts ($Pt–Ni/SiO_2$, $Pd–Ni/SiO_2$, and

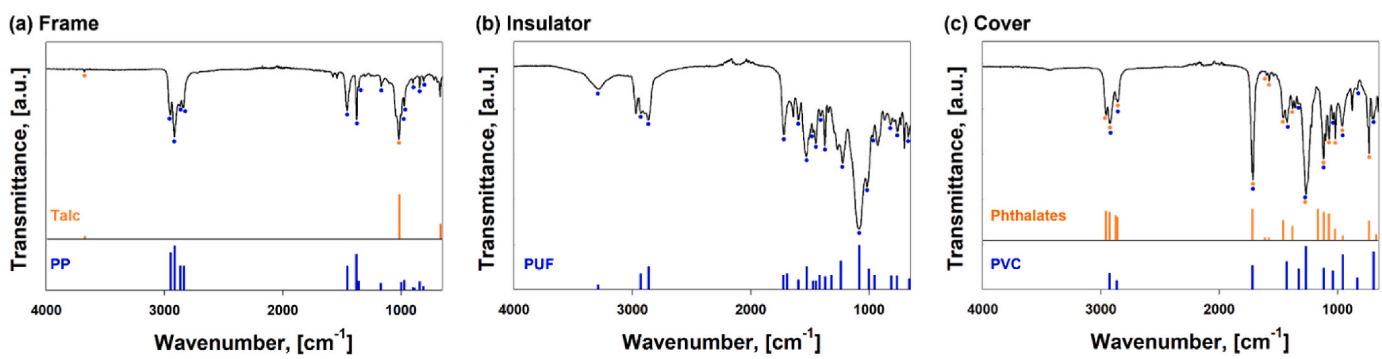


Fig. 2. FT-IR spectra of the (a) frame, (b) insulator, and (c) cover. Characteristic/reference peaks from the FT-IR spectra of PP, talc, PUF, PVC, and phthalates.

Rh–Ni/SiO_2), noble metal precursors (Pt, Pd, or Rh) were dissolved in DI water and impregnated onto the prepared Ni/SiO_2 catalyst. Identical calcination and reduction conditions were used for the dispersion of the noble metals on the Ni/SiO_2 catalyst to prepare the bimetallic catalysts. After the CatPyro of WDP, a temperature-programmed oxidation (TPO) test was performed using a TGA instrument under air. Temperature program for TGA was same with that in Section 2.2. The crystal structure of prepared catalysts were analyzed by using the X-ray diffractometer (XRD: Malvern Pananalytical Empyrean, UK) with Cu-Kα radiation source (λ = 1.5406 Å) at 40 kV and 30 mA. The XRD patterns were evaluated through step scanning in the 2θ range of 20–80° with intervals of 0.02°. Dispersion of metals on silica support was characterized by energy dispersive spectroscopy (EDX: Oxford Aztec One, UK) equipped with scanning electron microscope (SEM: EmCrafts CUBE 2, South Korea). Quantification of metal contents in silica support was performed by ICP-OES.

2.4. Reactor setups

Fixed-bed reactors were assembled for the WDP pyrolysis tests. Multiple experimental setups were employed to fundamentally understand the CO_2 effects during WDP pyrolysis. The single-stage pyrolysis study used a tubular furnace as the heating source, which covered a quartz tube reactor (600 mm long). The furnace temperature increased by 690 °C with a heating rate of 10 K min^{-1}; this temperature condition was identical to that used in the TGA test. One gram of WDP was placed at the midpoint of the reactor body. Multi-stage reactor with one more tubular furnace next to the single-stage setup was used for multi-stage pyrolysis. The latter heating zone was fixed at 600 °C, while the first heating zone had temperature increase. The length of the multi-stage reactor was 1000 mm. This setup was used to provide further thermal energy and increase the residence time of volatile matters (VMs) stemming from the WDP pyrolysis in the first heating zone. CatPyro was conducted using the same reactor set-up as that used for the multistage pyrolysis test. One gram of each catalyst was placed in the second heating unit. Pyrolysis tests were conducted in either N_2 or CO_2 environments at a gas flow rate of 200 mL min^{-1}. All the experiments were conducted least three times, and the errors were less than 3 %. Experimental setup is illustrated in Fig. S1.

2.5. Pyrogenic product analysis

Evolved pyrolysis oil and gas from WDP pyrolysis were analyzed quantitatively and qualitatively using two chromatography (GC) instruments. To capture condensable compounds (*i.e.,* pyrolysis oil), a cold trap (-40 °C) was connected to the outlet of the pyrolysis reactor. Volatile matters from pyrolysis of WDP were liquified in the solvent trap, filled with dichloromethane. The solution was then injected to GC mass spectrometer (GC/MS) to identify different HCs in pyrolysis oil. Non-condensable pyrolysis gas such as H_2, CO, CO_2 and C_{1-2} HCs was further flowed into the following online micro-GC as for gas identification and quantification. Detailed operation conditions for micro-GC and GC/MS are shown in Tables S1 and S2. Multipoint standard curves were drawn using a standard gas mixture provided by RIGAS (South Korea) for quantification and identification of pyrolysis gas through micro-GC. Multipoint standard curves for quantification of gaseous products obtained from Inficon Micro-GC was shown in Fig. S2. For identification of liquid compounds, GC/MS library and chromatogram of C_{8-40} alkane standards were also compared manually (Fig. S3). Solid content (plastic char + ash) remained after WDP pyrolysis was measured gravimetrically.

3. Results and discussion

3.1. Plastic types in WDP

As shown in Fig. 1, the WDP comprises three components (frame, insulator, and cover). Each component of WDP has a specific chemical structure (i.e., a monomer) [36]. To define the exact plastic types of the three components in the WDP, a series of FT-IR analyses were carried out. The FT-IR test enables the determination of the specific functional group in plastics [37]. The FTIR spectra of the frame, insulator, and cover are shown in Fig. 2.

As shown in Fig. 2(a), eight characteristic peaks (3679 cm^{-1}, 2950 cm^{-1}, 2915 cm^{-1}, 2838 cm^{-1}, 1455 cm^{-1}, 1377 cm^{-1}, 1019 cm^{-1}, and 675 cm^{-1}) were observed in the FT-IR spectrum of the frame. The peaks shown in Fig. 2(a) were observed in the FT-IR spectrum of PP. Elemental analysis also strongly supports the presence of PP (Table S1). The mass ratio of carbon to hydrogen in the frame sample was approximately equivalent to that of PP [38]. Thus, it was concluded that the frame in the WDP was composed of PP. The three characteristic peaks at 3679 cm^{-1}, 1019 cm^{-1}, and 675 cm^{-1} could also confirm the presence of talc in the frame [39]. Talc is used as a filler to impart desired physicochemical features to plastics by increasing their density while reducing their ductility [40,41].

Characteristic peaks at 2865 cm^{-1}, 1731 cm^{-1}, 1531 cm^{-1}, 1451 cm^{-1}, and 1223 cm^{-1} were observed in the FT-IR spectrum of the insulator in the WDP [Fig. 2(b)], which was indicative that the plastic type in the insulator was polyurethane foam (PUF) [42]. In addition, elemental analysis (Table S1) revealed that the insulator in the WDP was likely composed of PUF, considering the mass portion of nitrogen [43]. Considering the characteristic peaks at 1427 cm^{-1}, 1331 cm^{-1}, 1255 cm^{-1}, and 966 cm^{-1} shown in Fig. 2(c), it is pertinent that the cover in the WDP is composed of polyvinyl chloride (PVC) [42,44]. The characteristic peaks at 1725 cm^{-1}, 1601 cm^{-1}, 1581 cm^{-1}, 1267 cm^{-1}, and 741 cm^{-1} shown in Fig. 2(c) suggest that the cover contained phthalates [42,44]. Phthalates are widely used in plastics (particularly PVC) as effective additives to impart high plasticity [45,46]. Various phthalates are used to produce plastics [47]. Indeed, their identification by FT-IR analysis is difficult because of their similar chemical structures [48,

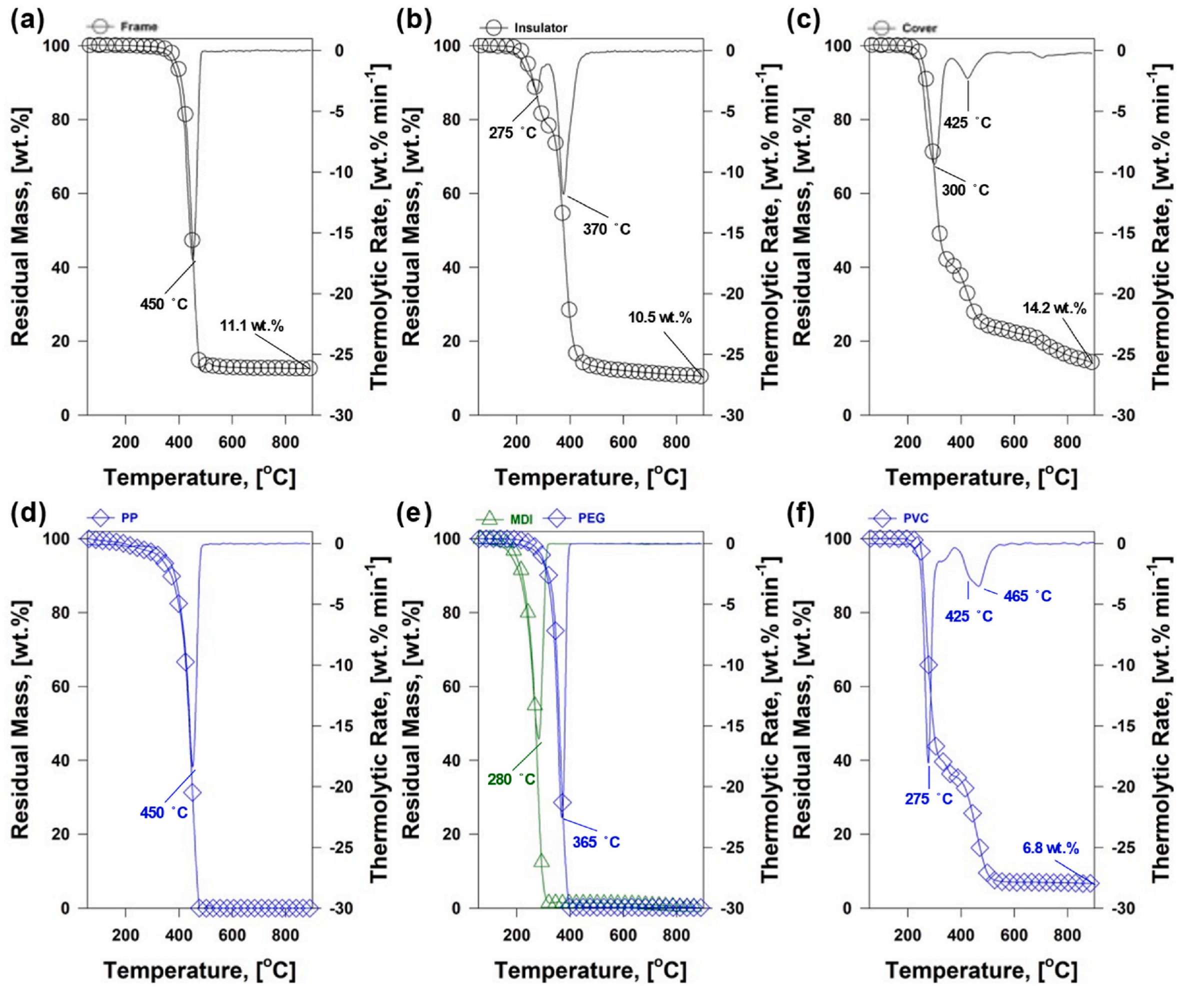


Fig. 3. Mass loss of (a) frame, (b) insulator (c) cover, (d) PP, (e) MDI/PEG, and (f) PVC and their thermolytic rates from N_2 condition: Primary y-axis indicates residual mass and secondary y-axis means thermolytic rate of each part.

49]. Phthalates in the three plastic components were chemically extracted using acetone to determine the exact types of phthalates in the WDP. Acetone was selected because of its high affinity for polar functional groups [50]. According to GC/MS analysis, three types of phthalates, [bis(2-ethylhexyl) phthalate, di(octan-2-yl) phthalate, and di(decan-2-yl) phthalate] were detected from cover and insulator.

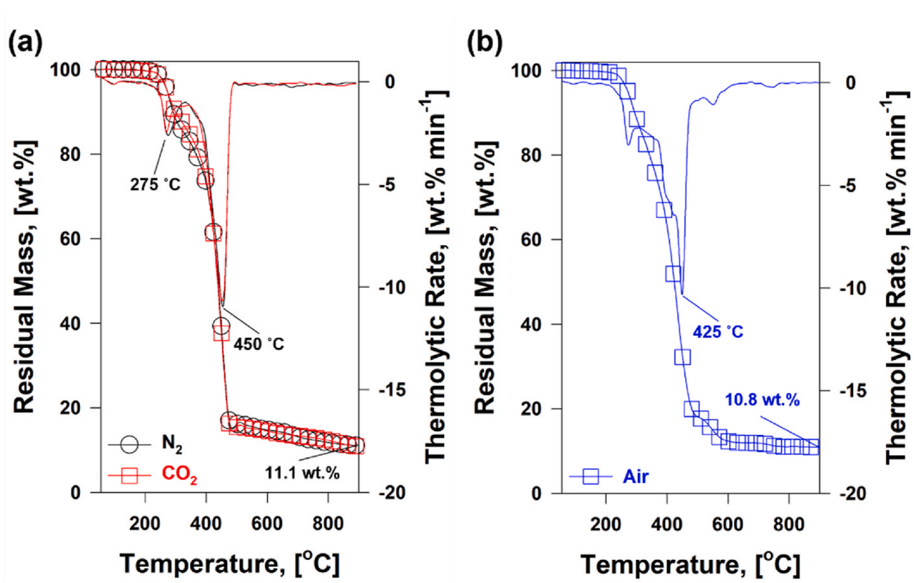


Fig. 4. Mass changes of WDP and its thermolytic rates under (a) N_2/CO_2 and (b) air.

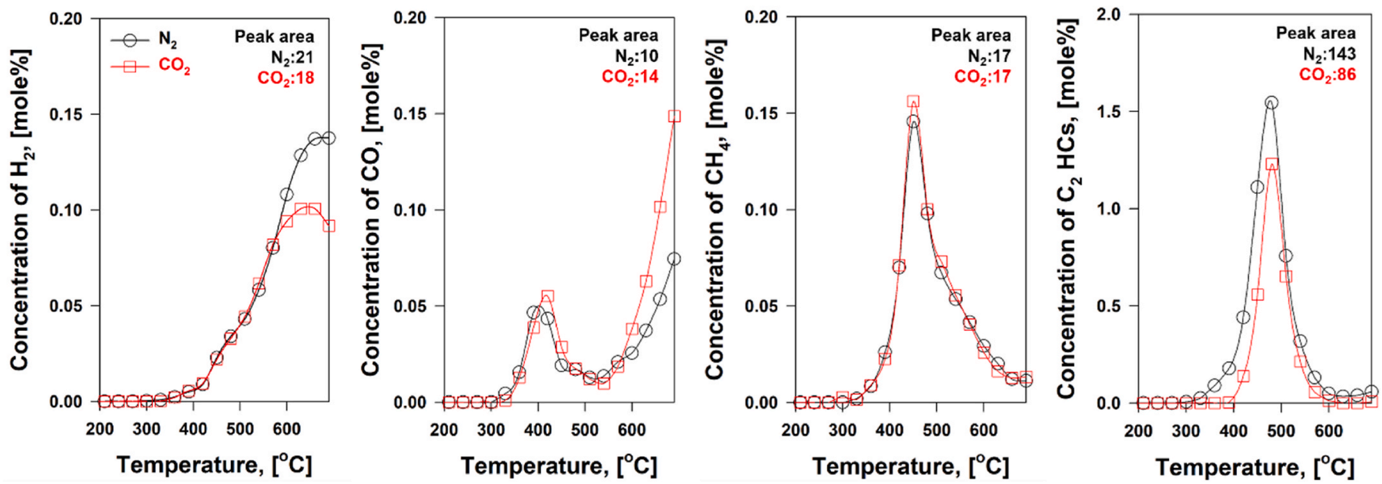


Fig. 5. Concentrations of pyrolysis gases from single-stage WDP pyrolysis under N_2 and CO_2 flowing gases. Peak areas of each pyrolysis gas produced under N_2 or CO_2 flowing gas are shown in the upper right.

Frame did not show phthalate (Fig. S4).

3.2. Thermolytic behaviors of WDP

To examine pyrolysis characteristics of WDP, it is of importance to monitor the thermal degradation of each plastic component in the WDP under different gases and temperature ranges. Indeed, the thermo-chemical process of WDP could be the co-pyrolysis of PP, PUF, and PVC. Therefore, TGA was carried out to characterize the trends in the thermolytic mass change (thermal stability) of the three WDP components. The mass changes of the frame, insulator, cover, PP, MDI/PEG, and PVC, and their thermolytic rates are shown in Fig. 3. PUF is composed of a monomer of MDI/PEG [51]. A substantial mass decrease (88.9 wt%) of frame was achieved at ≤ 500 °C [Fig. 3(a)]. This was likely due to the production of volatile matter (VM) in the frame. The temperature with the fastest thermolytic rate was 450 °C, which was identical to that of PP [Fig. 3(d)]. Given that the thermolysis of PP was completed at ≥ 500 °C, the final residual mass [11.1 wt% in Fig. 3(a)] at 900 °C was estimated as the talc mass. The presence of talc in the frame was confirmed by the FT-IR spectra shown in Fig. 2(a). The trends in the thermolytic mass change of PUF reflect stepwise thermal degradation [Fig. 3(b)]. The two characteristic DTG peaks at 275 °C and 370 °C were developed [Fig. 3(b)]. This is because PUF was prepared by the condensation polymerization of two monomers (diisocyanates and polyols) [52]. MDI/PEG monomers have been used to produce PUF [51]. The fastest thermolytic rates of MDI (280 °C) and PEG (365 °C) shown in Fig. 3(e) were in good agreement with the two distinctive peaks in the thermolytic rates of PUF. Nevertheless, the thermal degradation of the insulator occurred continuously up to 900 °C [Fig. 3(b)]. The presence of two benzene rings in the MDI monomer provides a favorable chance to form char via carbonization/dehydrogenation [53].

Two steps of mass change from the thermolysis of the cover were observed [Fig. 3(c)], and the fastest thermolytic rates at 300 °C and 425 °C were achieved. The temperatures of the fastest thermal degradation of PVC were like those of WDP cover at 300 °C and 425 °C [Fig. 3(f)]. The thermolysis of PVC was initiated by the dissociation of Cl/H via dehydrochlorination. Owing to the high electronegativity of Cl, the liberation of HCl from PVC can be achieved at ≥ 220 °C [54]. Subsequently, the liberation of hydrocarbons (HCs) is realized [55]. The second peaks in Fig. 4(c) and (f) were originated from thermal degradation of C–C backbone of PVC as other plastics showed same trends. Thus, trends in the thermolytic patterns of the cover could provide the real mass of PVC in the cover. However, a distinct thermolytic change in the cover was achieved by the liberation of phthalates. Specifically, the temperature window (220–360 °C) indicating the thermolytic mass change of cover by means of dehydrochlorination was wider than that (220–310 °C) of PVC. This is likely due to the dissociation of HCl and phthalates from the PVC polymeric backbone.

Fig. 4(a) shows the thermolytic mass change of the WDP under inert gas conditions. As shown in Fig. 1, the total mass of the frame (71.1 wt%) and cover (24.1 wt%) was much heavier than that of insulator (4.8 wt%). As such, the two characteristic peaks in the thermolytic rates of WDP at 275 °C and 450 °C could delineate the thermolysis of PVC and PP. As evidently demonstrated, the major mass change of WDP was completed at ≤ 500 °C. Subsequent carbonization in line with dehydrogenation contributes to the slight mass change from 500 °C to 900 °C.

To investigate the CO_2 effectiveness, TGA tests of the WDP under CO_2 conditions were also carried out. For comparison, the thermolytic mass decay of the WDP and its thermal degradation rate in the two gas environments are shown in Fig. 4(a). The thermal degradation patterns of the WDP under the two gas conditions were indistinguishable. Interestingly, the expected reaction by CO_2 reduction through the gasification reaction ($C_{(s)} + CO_{2(g)} \rightleftharpoons 2CO_{(g)}$) was not observed. The gasification reaction with CO_2 spontaneously happens at 710 °C and higher temperature ranges [56]. Thus, the absence of the gasification reaction can be ascribed to carbon scarcity. The final residual mass of the WDP in air [Fig. 4(b)] was 10.8 wt% of the WDP sample mass. The differences in the residual solid masses of the WDP under different gas flows were nearly negligible.

3.3. Pyrolysis of WDP in the CO_2 environment

Identical thermolytic trends for WDP were observed regardless of the types of atmospheric gases (Fig. 4). The identical mass change of the WDP under N_2/CO_2 indicates the no existence of additional chemical reaction of CO_2 with WDP. Indeed, non-homogeneous reactions/interactions of CO_2 with solid WDP can divert the thermolytic trends of WDP [57]. The TGA test does not provide a solid clue for the occurrence of chemical reaction between gaseous CO_2 and VMs produced from thermolysis of WDP. TGA apparatus only provides the gravitational change of the solid sample [58]. In an effort to seek the real occurrence of the CO_2-induced homogeneous reaction, the pyrolysis test of WDP under the two gas environments was performed from 35 °C to 690 °C. Evolution of representative gaseous products from WDP pyrolysis under CO_2 environment in reference to that under inert are shown in Fig. 5 to delineate the CO_2 effectiveness. Note that C_2-hydrocarbons (HCs) include C_2H_6, C_2H_4, and C_2H_2.

The concentration changes of the representative gaseous products

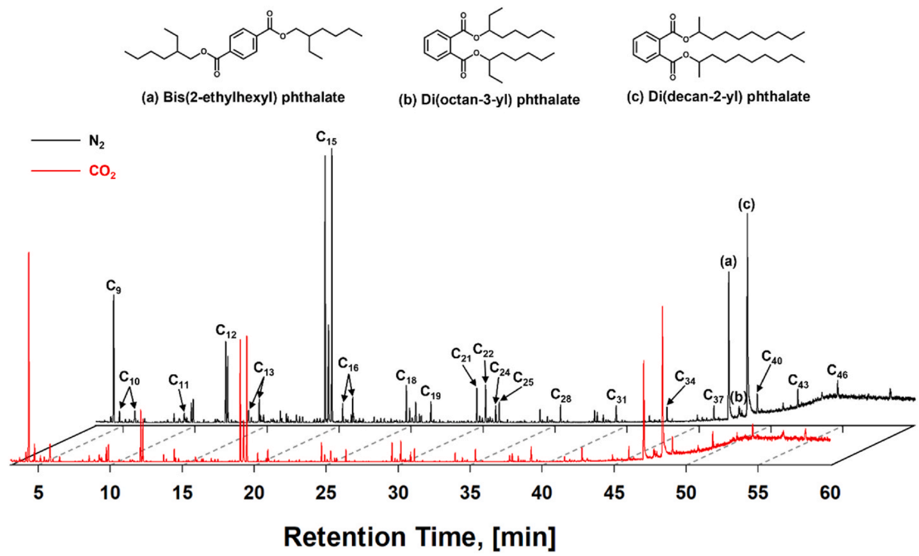


Fig. 6. Compositional matrices of wax-like/liquid HCs from the pyrolysis of WDP under the two gas environments.

from WDP pyrolysis were not substantially different ($H_2 \leq 0.14$ mol.%, $CO \leq 0.07$ mol.%, $CH_4 \leq 0.15$ mol.%, and C_2 HCs ≤ 1.55 mol.%). Nonetheless, the formation of C_2–HCs was distinctive from those of H_2, CO, and CH_4. The formations of H_2, CO, CH_4, and C_2–HCs were ascribed to the bond scissions of C–H, C–O, and C–C in the polymeric backbone of the frame, insulator, and cover, respectively [59–61]. The low concentrations of H_2, CO, CH_4, and C_2–HCs indicate that most carbons in the WDP were allocated to wax-like and/or liquid HCs rather than to gaseous pyrogenic products.

Overall, the concentration changes of H_2, CH_4, CO, and C_2 HCs, even in the CO_2 environment, were not noticeably distinguishable to those under inert gas conditions. However, CO concentration increase under the CO_2 environment was slightly shown at ≥ 570 °C. The CO concentration increase is in good agreement with previous work by the authors [62–64]. In detail, the mechanistic functionality of CO_2 at ≥ 550 °C led to the enhanced CO production by means of CO_2 reduction and the subsequent/simultaneous oxidation of VM (from the pyrolysis of the lignocellulosic biomass) [57]. The promoted CO production suggests that the CO_2 could contribute to the gas phase chemical reactions with VMs derived from WDP pyrolysis [33]. Nevertheless, the slightly enhanced CO concentration under the CO_2 atmosphere did not feature the mechanistic functionality of CO_2 in a fully transparent manner.

To examine the CO_2 effectiveness, the chemical compositions of pyrolysis oils (wax-like and liquid HCs mixture) from WDP pyrolysis were examined (Fig. 6). It was presumed that the overall yield of pyrolysis oils from WDP pyrolysis in a CO_2 environment could be lower than that from inert gas conditions. Specifically, the CO_2 functionality due to the homogeneous reactions could likely change the compositional matrix of the wax-like/liquid HCs. To confirm this, the compositional matrices of wax-like/liquid HCs from the pyrolysis of WDP under the two gas environments were visualized/compared, as shown in Fig. 6. The chemical species in the wax-like and liquid HCs and their peak areas are listed in Table S2.

Under inert gas conditions, the wax-like/liquid HCs formed from WDP comprised aliphatic HCs with wide-ranging carbon length ($C_{9\text{-}46}$). Aliphatic HCs in wax-like/liquid HCs mostly stemmed from PP pyrolysis [65]. The compositional matrix of wax-like/liquid HCs from the pyrolysis of WDP was similar to that of PP [65]. Considering the mass of the frame (71.1 wt%) in WDP, it was readily inferred that aliphatic HCs in wax-like/liquid HCs stemmed from PP pyrolysis in the frame. Aliphatic HCs in wax-like/liquid HCs from WDP could be used as fuels [gasoline ($C_{4\text{-}13}$), kerosene ($C_{6\text{-}26}$), and diesel ($C_{8\text{-}28}$)] [66]. The formation of

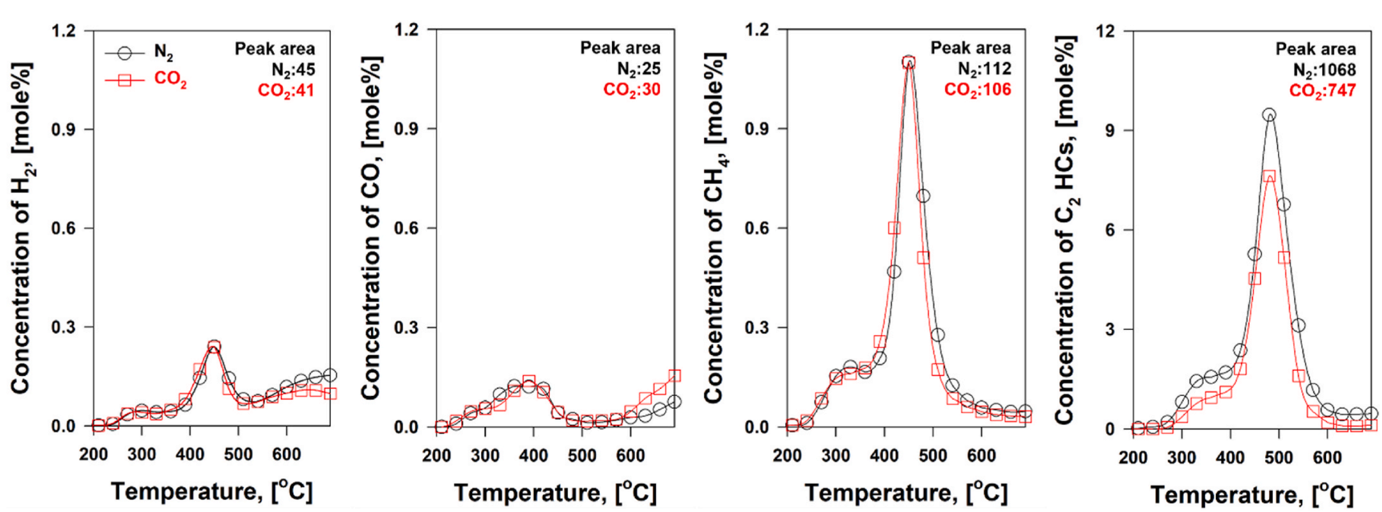


Fig. 7. Concentrations of pyrolysis gases from multi-stage WDP pyrolysis under N_2 and CO_2 flowing gases. Peak areas of each pyrolysis gas produced under N_2 or CO_2 flowing gas are shown in the upper right.

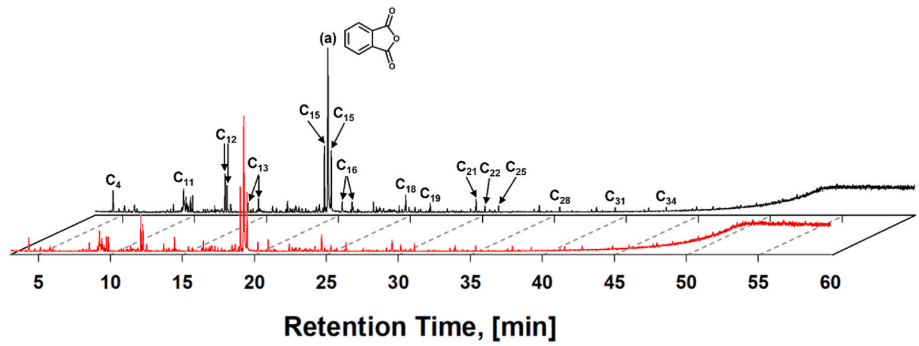


Fig. 8. Compositional matrixes of oils from multi-stage pyrolyses of WDP under the two gas environments.

chemical species from the insulator was negligible because of the low mass of the insulator (4.8 wt%). The chemical species formed by the insulator are not listed in Table S2. The three chemical species [denoted as (a), (b), and (c) in the chromatogram] were not formed by the pyrolysis of PP in the frame. These chemical species were phthalates [bis (2-ethylhexyl) phthalate, di(octan-2-yl) phthalate, and di(decan-2-yl) phthalate]. Phthalates originate from the pyrolysis of PVC on the cover [67]. Phthalates must be carefully managed because exposure to them via ingestion, inhalation, and dermal contact poses a potential hazard (hormonal disruption and cancer) [68].

In overall, quantity of pyrolysis oil produced from WDP pyrolysis under CO_2 environment was lower than that under inert gas conditions, while the shorter chain HCs were more produced under CO_2 condition. In detail, the quantity of the chemical species at $\leq$ 5.8 min from the CO_2 environment were larger than those from an inert gas condition. In contrast, the amount of the chemical species at $\geq$ 9.2 min from the CO_2 environment were diminished compared to those from an inert gas condition. In addition, the total peak areas of the chemical species in condensable HCs from the CO_2 environment decreased in comparison to those from inert conditions. The peak area patterns observed in Fig. 6 and Table S2 indicate that CO_2 promoted the chemical bonds cracking of Vs through the gas phase homogeneous reactions. In addition, the promoted chemical bonds cracking induced by CO_2 could provide a strategic means for making the pyrolysis process more reliable. Notably, coke deposits in pipelines disturb the flow of fluids in pyrolysis systems [69].

3.4. Multi-stage pyrolysis of WDP in the CO_2 environment

Increased chemical bonds cracking of VMs induced by CO_2 through a homogeneous reaction was observed. Nonetheless, CO_2 reduction to CO and simultaneous oxidation of VM by means of a homogeneous reaction were not achieved, as shown in Fig. 5. In detail, the enhanced formation of CO was observed at $\geq$ 570 °C, but its enhancement was not substantial. In addition, the pyrogenic products of the liberation of VM from the WDP mostly evolved at $\leq$ 500 °C (Fig. 4). To provide favorable conditions for the reaction between gaseous CO_2 and VM, the experimental setup was modified. Specifically, the additional thermal energy was provided to expedite CO_2 reduction and VM oxidation into CO production at 600 °C. The trends in the evolution of the major gaseous products (H_2, CO, CH_4, and C_2–HCs) from the pyrolysis of WDP in the two gas environments are shown in Fig. 7.

As shown in Fig. 7, the expected gas-phase reactions (CO_2 reduction and VM oxidation to CO) did not develop. In addition, gas production (H_2, CO, CH_4, and C_2–HCs) in the two gas environments was nearly identical. This observation suggests that the reaction kinetics for CO_2 reduction to CO and simultaneous oxidation of the pyrogenic products are not fast. Interestingly, the formation of CH_4 and C_2–HCs from the multi-stage pyrolysis of WDP (compared to single-stage pyrolysis) was substantially enhanced. It was concluded that C–C bond scission mainly occurred. This subsequently led to the reduced formation of wax-like liquid HCs from the pyrolysis of WDP in the two gas environments. To confirm this, the identification of oil components from the multi-stage pyrolysis of WDP under N_2 and CO_2 atmospheres was visualized and compared in Fig. 8. The chemical species present in the pyrolysis oils and

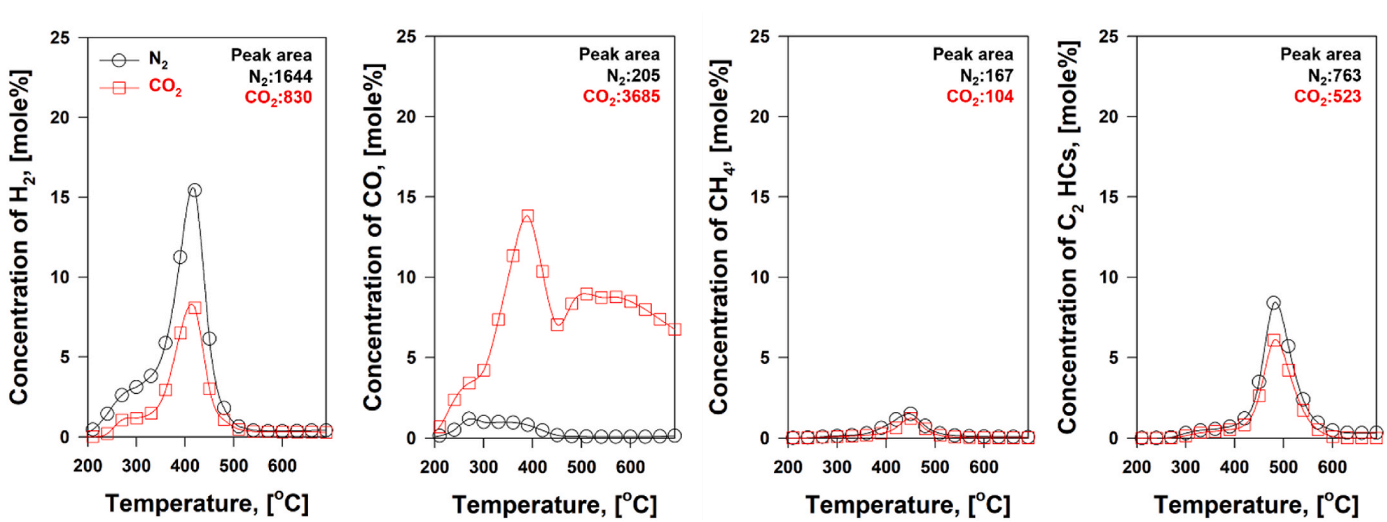
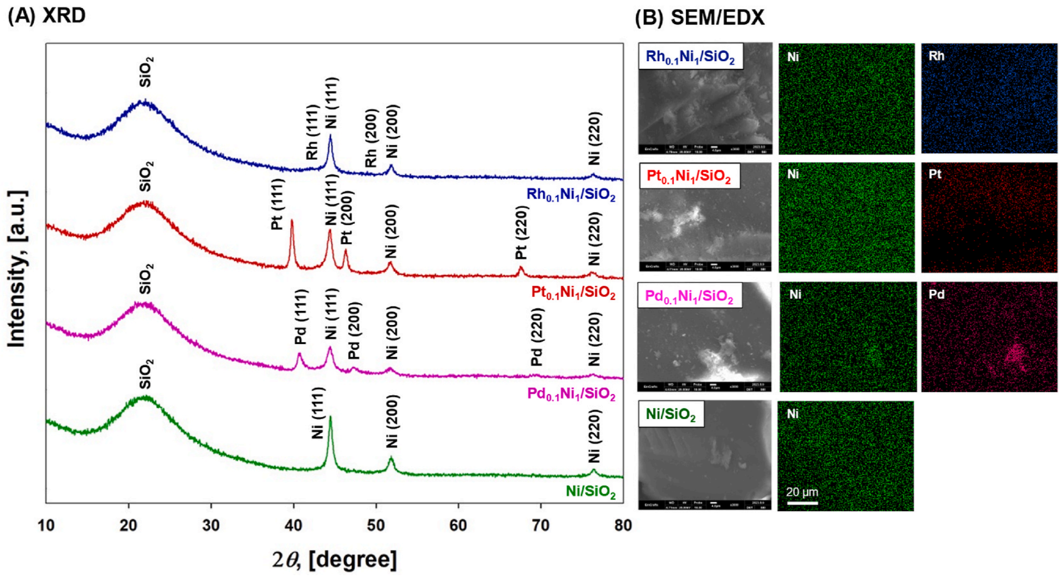


Fig. 9. Concentrations of pyrolysis gases from CatPyro of WDP pyrolysis under N_2 and CO_2 flowing gases in the presence of Ni/SiO_2 catalyst. Peak areas of each pyrolysis gas produced under N_2 or CO_2 flowing gas are shown in the upper right.



(C) ICP-OES

Catalyst	Metal loading on SiO_2 support			
	Ni loading (wt.%)	Ni loading (mole%)	Noble metal loading (mole%)	Noble metal to Ni ratio (by mole)
Ni/SiO_2	4.96	5.07	-	-
$Pd_{0.1}Ni_1/SiO_2$	4.90	4.99	0.51	0.10
$Pt_{0.1}Ni_1/SiO_2$	4.81	4.90	0.50	0.10
$Rh_{0.1}Ni_1/SiO_2$	4.86	4.95	0.51	0.10

Fig. 10. (a) XRD patterns, (b) SEM images, and EDX mapping of Ni/SiO_2, $Pd_{0.1}Ni_1/SiO_2$, $Pt_{0.1}Ni_1/SiO_2$, and $Rh_{0.1}Ni_1/SiO_2$ catalysts, and (c) metal loading on SiO_2 support determined by ICP-OES analysis.

their corresponding peak areas are listed in Table S3.

As shown in Fig. 8 and Table S3, the amount of chemical species in the oil from the multi-stage pyrolysis of the WDP (in reference to single-stage pyrolysis) was substantially reduced. This is closely related to the enhanced generation of CH_4 and C_2–HCs (Fig. 7). Specifically, the enhanced bond scissions of C–C under the circumstance of additional thermal energy led to a reduced amount of chemical species in the oil. In addition, most phthalates are converted into phthalic anhydrides [70]. Thus, it was concluded that the reaction kinetics for CO_2 reduction to CO and the simultaneous oxidation of the pyrogenic products were not fast.

To expedite the homogeneous reaction (CO_2 reduction to CO and simultaneous oxidation of the pyrogenic products), CatPyro of WDP over Ni/SiO_2 was carried out. For comparison, CatPyro of WDP over Ni/SiO_2 was performed in two gas environments. The trends in the evolution of the major gaseous products (H_2, CO, CH_4, and C_2–HCs) from the pyrolysis of WDP in the two gas environments are shown in Fig. 9.

As shown in Fig. 9, the formation of H_2 under inert gas conditions was notably enhanced. The peak area of H_2 from CatPyro under inert gas conditions was 36 times larger than that of the multi-stage system. This suggests that the catalytic capability of Ni/SiO_2 enhanced the formation of H_2. In a CO_2 environment, the formation of CO was significantly enhanced. Indeed, the peak area of CO from CatPyro in the WDP in the CO_2 environment was 122 times larger than that in the multistage environment. Such enhanced formation of CO could be key for the CO_2 reduction to CO and the simultaneous oxidation of the pyrogenic products through a homogeneous reaction. The conversion of WDP into the smallest molecular units, such as CO, offers an innovative means of neutralizing toxic chemicals. In other words, the enhanced formation of CO by CO_2 offers a strategy for the breakdown of phthalate derivatives.

The enhanced formation of CO (Fig. 9) in the homogeneous reaction could be described by two mechanistic routes. First, reverse water-gas-shift (rWGS) reaction ($H_2+CO_2 \rightleftharpoons CO + H_2O$) could contribute to the enhanced formation of CO [71]. Second, the CO_2 reduction to CO and the simultaneous oxidation of the pyrogenic products by means of a homogeneous reaction could also contribute to the enhanced generation of CO. Nonetheless, CO enhancement by the rWGS reaction is unlikely because the rWGS results in an unimolecular stoichiometric reaction [72]. In other words, CO enhancement by means of the rWGS reaction could not be realized, considering the peak areas of CO and H_2 in Fig. 9. Thus, it was concluded that the CO enhancement was mainly ascribed to CO_2 reduction to CO and the simultaneous oxidation of the pyrogenic products by means of a homogeneous reaction. To accelerate the kinetics of the homogeneous reaction, bimetallic Ni catalysts were tested with noble metals. Bimetallic Ni catalysts were prepared using Pd, Pt, and Rh. To impart an identical number of active sites, the amounts of Pd, Pt, and Rh were controlled at a molar ratio of 0.1. Three bimetallic catalysts ($Pd_{0.1}Ni_1/SiO_2$, $Pt_{0.1}Ni_1/SiO_2$, and $Rh_{0.1}Ni_1/SiO_2$) were prepared, and the catalyst characterizations were performed using XRD, SEM/EDX, and ICP-OES analyses. XRD results confirmed the presence of both Ni and noble metals on silica support (Fig. 10(a)). SEM image and EDX mapping showed uniform dispersion of Ni and noble metals on the support (Fig. 10(b)). For quantification of metal loadings, ICP-OES was performed (Fig. 10(c)). ICP-OES results confirmed that 5 wt% of Ni was loaded on SiO_2 support, and molar ratio of noble metal to Ni on SiO_2 was 1: 10.

Using the prepared bimetallic catalysts, CatPyro of WDP was performed. The trends in the evolution of the major gaseous products (H_2, CO, CH_4, and C_2–HCs) from the CatPyro of WDP over $Pd_{0.1}Ni_1/SiO_2$, $Pt_{0.1}Ni_1/SiO_2$, and $Rh_{0.1}Ni_1/SiO_2$ in the two gas environments are shown in Fig. 11. As shown in Fig. 11, the trends in the gaseous products from the CatPyro of the WDP over $Pd_{0.1}Ni_1/SiO_2$, $Pt_{0.1}Ni_1/SiO_2$, and $Rh_{0.1}Ni_1/SiO_2$ were different. $Rh_{0.1}Ni_1/SiO_2$ was the most effective in expediting the CO_2 reduction into CO and the simultaneous oxidation of the pyrogenic products by means of a homogeneous reaction considering the total peak areas of H_2, CO, CH_4, and C_2–HCs.

For an in-depth study, the molar ratio of Rh varied from 0.05 to 0.2.

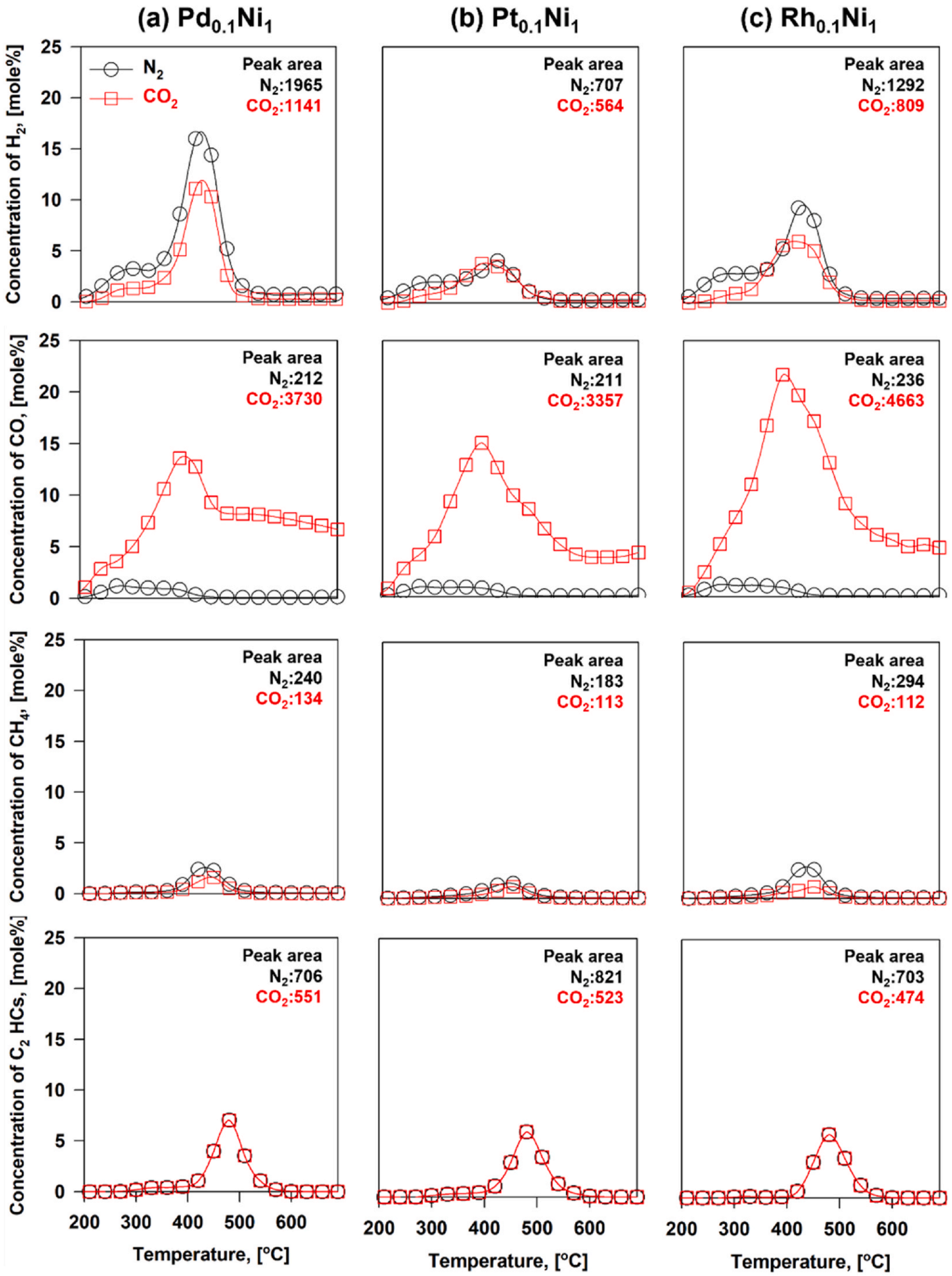


Fig. 11. Trends in the evolution of the major gaseous products (H_2, CO, CH_4, and C_2–HCs) from the CatPyro of WDP over $Pd_{0.1}Ni_1/SiO_2$, $Pt_{0.1}Ni_1/SiO_2$, and $Rh_{0.1}Ni_1/SiO_2$ under the two different flowing gases (N_2 and CO_2). Peak areas of each pyrolysis gas produced under N_2 or CO_2 flowing gas are shown in the upper right.

Specifically, CatPyro tests of the WDP over $Rh_{0.05}Ni_1/SiO_2$, $Rh_{0.1}Ni_1/SiO_2$, and $Rh_{0.2}Ni_1/SiO_2$ were conducted in a CO_2 environment. To evaluate the catalytic capability, the relative mole percentages of the major gaseous products from the CatPyro of WDP over $Rh_{0.05}Ni_1/SiO_2$, $Rh_{0.1}Ni_1/SiO_2$, and $Rh_{0.2}Ni_1/SiO_2$ were compared, as shown in Fig. 12. CO enhancement was proportional to Rh loading. In contrast, the formation of C_2–HCs was inversely proportional to Rh loading because the reaction kinetics governing CO_2 reduction into CO and the oxidation of the pyrogenic products by means of the homogeneous reaction were accelerated. In other words, more pyrogenic products were partially oxidized by CO_2 through a homogeneous reaction.

The more production of syngas also indicates that plastic oil production was reduced, because plastic oil was converted into syngas. To confirm the estimation, plastic oil collected after CatPyro of WDP was compared in Table S4. $Rh_{0.1}Ni1/SiO_2$ catalyst more transformed plastic oil into syngas than Ni/SiO_2.

The TPO tests strongly supported the catalytically enhanced CO_2 reduction to CO and oxidation of the pyrogenic products. TPO tests determine the amount of (hydro)carbon deposition on catalyst by monitoring the residual mass difference between unused and used catalysts. When the (hydro)carbon was deposited on the catalyst, oxidation at high temperature with air flow decreases the residual mass of catalyst.

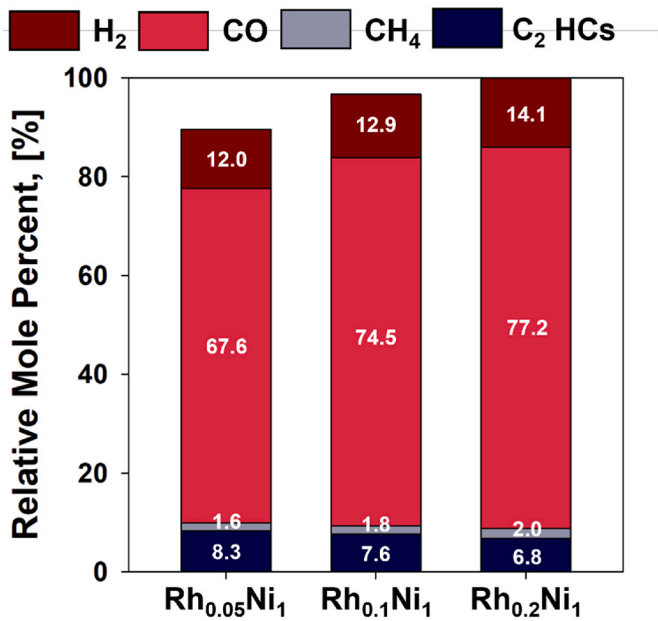


Fig. 12. Relative mole percentages of the major gaseous products from the CatPyro of WDP over (a) $Rh_{0.05}Ni_1/SiO_2$, (b) $Rh_{0.1}Ni_1/SiO_2$, and (c) $Rh_{0.2}Ni_1/SiO_2$ catalysts under the CO_2 environment.

As such, more mass decrease means more carbon deposition. When $Rh_{0.1}Ni_1/SiO_2$ catalyst was used, less carbon deposition was shown than Ni monometallic catalyst as shown in Fig. 13. Under CO_2 condition, carbon deposition was substantially suppressed, because CO_2 likely converted hydrocarbons into CO over catalyst surface. $Rh_{0.1}Ni_1/SiO_2$ bimetallic catalyst further suppressed carbon deposition than Ni catalyst under CO_2-assisted pyrolysis. As aforementioned, homogeneous gas phase reactions are likely attributed to the suppression of coke deposition because volatile hydrocarbons were converted into CO under CO_2 condition instead of coke formations.

In this study, the effects of CO_2 and Ni bimetallic catalysts on syngas production and catalyst stability during plastic waste pyrolysis were investigated. In the presence Ni catalyst, plastic oil is converted into syngas and $C_{1\text{-}2}$ HCs, but carbon deposition on Ni catalyst was challenging. Introduction of CO_2 suppressed carbon deposition on Ni catalyst with additional CO formation. When bimetallic catalysts ($Pd_{0.1}Ni_1/SiO_2$, $Pt_{0.1}Ni_1/SiO_2$, and $Rh_{0.1}Ni_1/SiO_2$) were used, more syngas production and lower carbon deposition were found. $Rh_{0.1}Ni_1/SiO_2$ showed the best performance in syngas production during CO_2-assisted catalytic pyrolysis. This work revealed the effects of CO_2 and bimetallic catalysts, but several studies must be done to apply the founding of this study to practical pyrolysis systems. The quantitative analysis of CO_2 contribution to plastic conversion to CO will be of importance to understand the detailed reaction mechanisms over pyrolysis catalysts. Optimization of the reactor setups and mass balance construction will be required. Comparison with other Ni-based catalysts in the similar reactor setups can help to understand the superiority of bimetallic catalysts [73]. Evaluation of economic feasibility of plastic waste pyrolysis under the CO_2 condition with different catalysts and various operating conditions need to be considered, which could lead to development of the most efficient advanced pyrolysis processes for disposing and valorizing plastic wastes [74].

4. Conclusions

This study proposes a thermochemical disposal platform for plastic waste derived from ELVs. To convert heterogeneous plastic and chemical mixtures (e.g., PP, PUF, PVC, phthalate, talc, *etc.*) from ELVs into value-added products during disposal, catalytic pyrolysis was used to produce syngas. To develop a greener thermochemical platform for waste mixture conversion, CO_2 was employed as a green reaction medium for the conversion of CO_2 and volatile matter from plastics into CO and H_2. Catalytic pyrolysis converts the complicated hydrocarbon species in plastic oil into syngas. Ni/SiO_2 is effective; however, carbon deposition is problematic. In addition to using CO_2 as a carrier gas, syngas production more than doubled through catalytic pyrolysis in comparison to that under N_2 conditions. In addition, carbon deposition decreased by 80 %. CO_2 also contributes significantly to the conversion of volatile hydrocarbons from plastic to CO through a gas-phase reaction between CO_2 and volatile compounds. To further suppress catalyst deactivation and enhance syngas production, bimetallic catalysts were prepared by adding noble metals such as Pt, Pd, and Rh. Among these, the Rh-supported bimetallic catalyst ($Rh_{0.1}Ni_1/SiO_2$) was the most effective. Syngas formation was enhanced by 12 % and carbon deposition decreased by half during catalytic pyrolysis with $Rh_{0.1}Ni_1/SiO_2$ compared to catalytic pyrolysis with Ni/SiO_2 (CO_2 conditions).

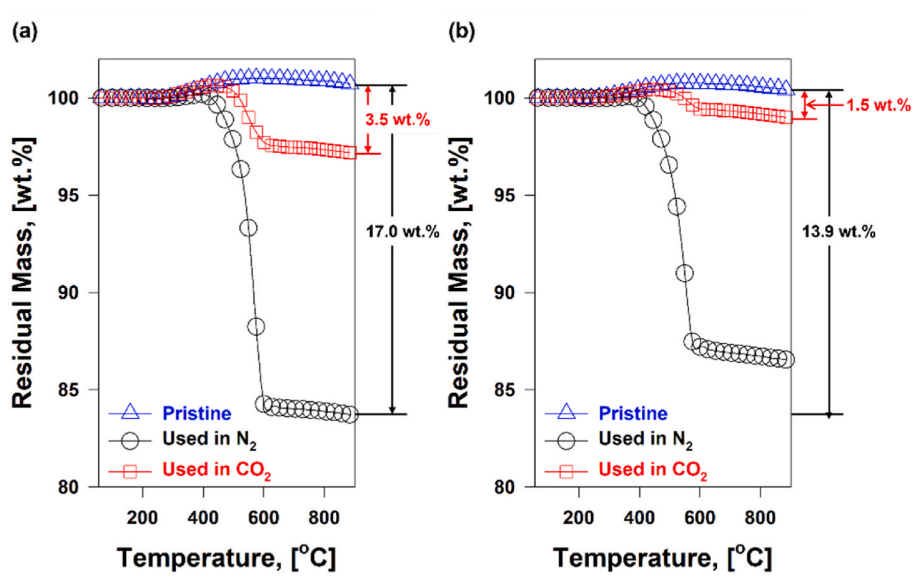


Fig. 13. TPO tests of (a) Ni/SiO_2 and (b) $Rh_{0.1}Ni_1/SiO_2$ from the two gas environments.

CRediT authorship contribution statement

Jung-Hun Kim: Data curation, Formal analysis, Investigation, Writing – original draft. **Sungyup Jung:** Formal analysis, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Tae-woo Lee:** Data curation, Formal analysis, Writing – review & editing. **Yiu Fai Tsang:** Formal analysis, Investigation, Resources, Writing – review & editing. **Eilhann E. Kwon:** Conceptualization, Data curation, Investigation, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.energy.2023.130136.

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